

- 2 (a) (at very low pressure,) gas occupies very large volume / particles are further apart
OR larger volume of gas means (finite) volume of particles becomes negligible [M1]

(large distance between particles) means any (finite) forces between them become negligible [A1]

- (b) (i) Show that pV at B is not equal to pV at C. [1]

- (ii) WD from C to A = 0 [1]

$$\begin{aligned} \text{WD from A to B} &= -2.6 \times 10^5 \times (3.8 - 2.3) \times 10^{-3} \\ &= -390 \text{ J} \end{aligned} \quad [1]$$

(iii)

Process	Heat supplied, Q / J	Work done on gas, W / J	Change in internal energy, ΔU / J
A to B	1370	- 390 ^[0, from bii]	980 ^[1]
B to C	0	550	550 ^[1]
C to A	- 1530 ^[2]	0 ^[0, from bii]	- 1530 ^[2]

[2, ecf]

- (iv) Net work done on gas = $-550 + 390 = -160 \text{ J}$

Heat absorbed = 1530

[1m for either net WD or Q, ecf from (iii)]

$$\text{Efficiency} = 160 / 1530 = 0.105 \text{ or } 10.5\% \quad [1]$$

3 (a) Oscillatory motion where the acceleration is proportional to the displacement from the